Appendix 1: ID Alignment

Instruction Manual

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I.Alignment

The effective ID is 11 bits for standard frames and 29 bits for extended frames. When a 32-bit unsigned plastic is used to store the ID, there are two ways of storing it: right-aligned (direct ID method) and left-aligned (SJA1000/register method). The alignment methods covered in this USBCAN device interface function library are all compatible with ZLG's USBCAN family of device interface function libraries!

1.1 Right Alignment

unsigned int (bit31～bit0)																																
Height																															Low	Location
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Bit
																					10	9	8	7	6	5	4	3	2	1	0	Standard Frame
			28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Extended Frame

1.2 Left Alignment

Left-aligned, the highest valid bit of the ID, ID.10 (standard frame), ID.28 (extended frame), is aligned with Bit.31. This is shown in the table below:

unsigned int (bit31～bit0)																																
Height																															Low	Location
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Bit
10	9	8	7	6	5	4	3	2	1	0																						Standard Frame
28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0				Extended Frame

II.ID Right-Aligned Approach (Direct ID Approach)

The CAN message ID is represented in the interface function library as unsigned plastic data with a total of 32 bits, of which 11 bits are valid ID for standard frames and 29 bits for extended frames, and is right-aligned (direct ID method), i.e., the lowest valid bit of the ID, ID.0, is aligned with the Bit.0 bit. This is shown in the table below:

ID(bit31～bit0)																																
Height																															Low	Location
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Bit
																					10	9	8	7	6	5	4	3	2	1	0	Standard Frame
			28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Extended Frame

Note: In the above table, the corresponding bit of the cell with dark background indicates the valid bit of ID, and the lowest valid bit of ID, ID.0, is aligned with Bit.0.

Example:

1. standard frame ID: If the standard frame ID is 00 00 01 23 (HEX), the standard frame ID is 11 bits valid, so the actual ID value is 00 00 01 23H.

2. Extended Frame ID: If the Extended Frame ID is 1F 01 02 03 (HEX), the Extended Frame ID is 29 bits valid, so the actual ID value is 1F 01 02 03H.

III.AccCode/AccMask Left-Aligned Mode (Register Mode)

AccCode (Filter Acceptance Code)/AccMask (Filter Mask Code) is represented in the interface function library by unsigned shaped data with a total of 32 bits, of which the effective ID for standard frames is 11 bits and for extended frames is 29 bits, using a left-aligned method (register method), i.e., the highest effective bits of the ID ID ID.10 (standard frames), ID.28 (extended frames) are aligned with Bit.31. This is shown in the table below:

AccCode/AccMask (bit31～bit0)																																
Height																															Low	Location
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Bit
10	9	8	7	6	5	4	3	2	1	0																						Standard Frame
28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0				Extended Frame

Note: In the above table, the corresponding bit of the cell with dark background indicates the valid bit of ID, and the highest bit of ID is aligned with the Bit31 bit.

Example:

1. Standard Frame ID: If the standard frame ID is 00 00 01 23 (HEX), the standard frame ID is 11 bits valid, so the actual ID value should be shifted left by 21 bits to get the AccCode/AccMask value 24 60 00 00H.

2. Extended Frame ID: If the extended frame ID is 1F 01 02 03 (HEX), the extended frame ID is 29 bits valid, so the actual ID value should be shifted left by 3 bits to get the AccCode/AccMask value F8 08 10 18H.